Surgical Innovation Discovery Course: Utilizing Surgical Trainee Agile Innovation and Empowerment (STAIR) Framework to Promote Innovation Amongst Surgical Residents

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INTRODUCTION: Innovation education programs can help guide and empower surgeons and surgical trainees through the many steps to analyze and de-risk novel ideas to impact healthcare or care delivery. Providing quality programming that is conducive to a surgeon's busy schedule is an ongoing challenge, but even more so for a surgical trainees' schedule. Through a needs assessment and applying lean principles, the "Surgical Trainee Agile Innovation and empoweRment" (STAIR) framework was created to encourage surgical trainees to participate in innovation programming.

OBJECTIVE: Numerous works have explored the design and development of learning experiences to support physician innovators. However, there is a lack of evidence supporting frameworks that contribute to innovation empowerment for surgical trainees as a complement to the demands of clinical practice. The following catalogs the systematic design and development of a novel innovation and discovery agile framework tailored to surgical residents with original innovation ideas.

DESIGN AND SETTING: The course designed to implement the "Surgical Trainee Agile Innovation and empoweRment" (STAIR) is funded through the Department of Surgery at the University of Michigan. It is an 8-week team-based innovation course designed for surgical residents to learn and apply fundamental innovation concepts to clinical problems.

PARTICIPANTS AND RESULTS: Surgical trainees in all programs at the host institution are all eligible to participate. A total of four innovation teams were selected to participate in inaugural course.

CONCLUSIONS: The novel instructional design and development of this course encouraged participation in surgical innovation programming. The course aided the development of innovative ideas and inspired interest in innovation among surgical trainees. The STAIR framework is exemplary for inspiring and facilitating innovation among surgical trainees, enhancing the professional experience. (J Surg Ed 81:1089−1093. © 2024 Association of Program Directors in Surgery. Published by Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.)

KEYWORDS: surgery, innovation education, medical devices, prototyping, agile methodology

AGCME COMPETENCIES: Interpersonal and Communication Skills, Professionalism, Practice-Based Learning and Improvement, Systems-Based Practice

INTRODUCTION

The Center for Surgical Innovation (CSI) aims to support physician-innovators aiding the development of novel healthcare and surgical related innovations through tailored and adapted innovation programming, mentorship, and resources. We recognize the need to create a unique course that accounts for the specific needs (e.g., work limitations and scheduling) of surgical trainees.

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1931-7204/$30.00 https://doi.org/10.1016/j.jsurg.2024.04.009

A Tailored and Adapted Innovation Course for Surgical Trainees

Through qualitative interviews with surgical trainees, findings supported a strong interest to participate in a structured team-based innovation program designed to increase efficiency and maximize resources (e.g., access to engineers, funding, and tangible prototypes). We crafted the Surgical Trainee Agile Innovation and empoweRment (STAIR) framework to support trainee-led innovations, applied in an 8-week team-based course. Surgical trainees submitted applications and served as Project Investigator (PI). A total of ten applications were submitted spanning seven surgical programs. Applications were assessed based on an unmet clinical need, including scope and frequency of problem, potential solution, prototyping feasibility, and recruitment of team members (Table 1).

TABLE 1. Surgical Innovation Discovery Course Application

The Surgical Innovation Discovery Course Application Questions:

1. Provide your first & last name.
2. Provide university email.
3. Provide residency program and training year.
4. What is the unmet surgical need or problem you would like to explore? Also describe the current standard of care.
5. What is your solution to this problem?
6. If your idea is successful, what impact do you believe your innovation will have on the field of academic surgery or patient outcomes?
7. List existing team members who will help you with this project. Indicate if you need help recruiting additional team members.
8. What best describes the stage of your current prototype of your solution?
   a. 1+ iterations of low-fidelity prototype already created
   b. Sketched out design but nothing built
   c. I have an idea in my head but I haven't sketched or created anything just yet
   d. I don't know where to begin or how to start this process
   e. Other
9. What do you envision your prototype artifact to be?
   a. Sketch and diagrams
   b. 3D printed model
   c. Physical Model
   d. Wireframe/Digital App
   e. Video
   f. Other
10. A starting course date will be finalized after all teams are selected. The course will meet in person for the first and last session. This course will require active participation and regular team check-ins throughout the 8-week program. Do you have full intentions of being an active participant in this program and dedicating time to work with your team on your identified project?
    a. Yes
    b. No
    c. Maybe

The course was facilitated by the CSI leadership team consisting of two directors, one fellow, and one biomedical engineer who worked in tandem to coach teams, remove hurdles, and ensure progress. Four applications were selected to participate in the course. All teams were supported by a team consisting of at least one but no more than three student volunteers. CSI leadership set expectations that PIs should demonstrate leadership and be responsive to team needs within 24-hours. All team members were expected to contribute 2-4 hours per week on their project and be reasonably accessible.

Applying the STAIR Framework

The STAIR framework (Fig. 1) begins with (1) *Observation & Innovation*, where surgical trainees apply to participate in the course with a pre-identified problem and a potential solution. Team 1, "Novel Surgical Light" co-led by General Surgery PGY-2 and PGY-7; Team 2, "New Device for Post Operative Pain Control" led by a Neurosurgery PGY-4; Team 3, "Endotracheal Tube Redesign" co-led by Otolaryngology PGY-4 and PGY-6; Team 4, "ECMO Blood clot reduction" led by a General Surgery PGY-5. Five out of the six surgical trainees participated in the course while on active clinical rotations. One surgical trainee was on their dedicated research time and did not have clinical obligations. Examples of how Team 1 utilized the STAIR framework will be highlighted throughout this paper.

Fig. 1 illustrates the STAIR framework to empower early-stage innovation. Through (1) *Observation & Innovation*, residents observe and identify a problem within their clinical duties that merits improvement. During (2) *Preparation*, surgical trainees brainstorm actionable plans to solve their identified problem and form teams to assist in their solution. Teams co-create interview guides, customer discovery strategies, and formulate prototype supplies. Teams then begin (3) *Prototyping* their design to model a physical representation of their solution. Once a prototype is developed, teams being their (4) *Daily Scrum & Customer Discovery* for one week. Each day during the week, the team shares the responsibility of conducting interviews and collecting feedback related to their identified problem, solution, and prototype. The team briefly meets or communicates each day to report progress, obstacles, and assigns the next day's tasks. Once complete, teams reconvene to (5) *Refine* their solution based on analyzing data collected during the sprint. Teams may choose to reiterate their prototype and complete tasks (3-5) or conclude they have met their (6) *Definition of Done*.

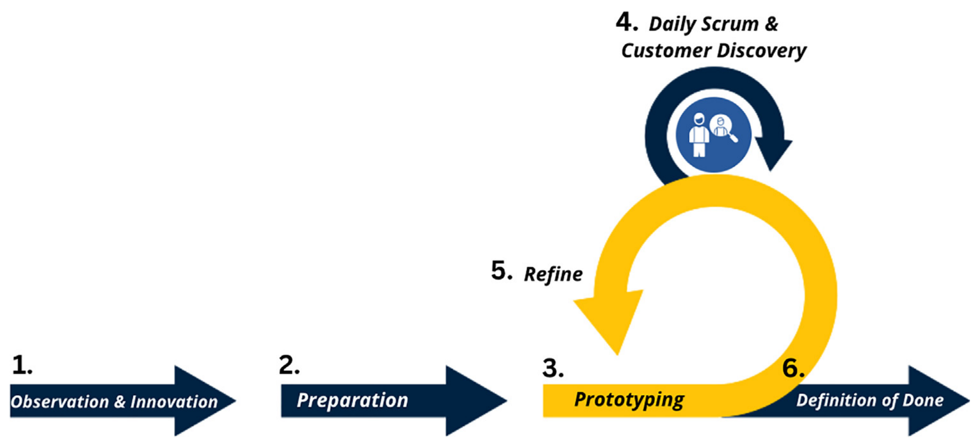

FIGURE 1. Surgical trainee agile innovation and empoweRment framework.

Team 1: "Novel Surgical Light," PIs expressed that during surgical procedures located in tight enclosed anatomical spaces with inadequate retraction or illumination of critical anatomy causes difficulty for surgeons and assistants to properly see and carry out given portions of operations. Current lighting options such as overhead lights, headlamps, or tethered lighted retractors are not useful during tight enclosed procedures.

Moving to (2) *Preparation*, the CSI helped surgical trainees recruit team members. Students were recruited by word of mouth and email. Surgical trainees reviewed and selected volunteers to join their team. Research teams consisted of a combination of M1-M3 medical students and graduate engineering students at the University of Michigan. Team 1 recruited one M3 student with an engineering background capable of building their prototype.

(2) *Preparation* also consisted of all four teams attending a 2-hour in person kickoff workshop together (Table 2). PIs selected the date of the workshop 90-days in advance based on clinical schedules and team member availability. During the workshop, teams established rapport, trust, and established their asynchronous work and communication plan for their participation in the course. Teams co-created expectations and identified responsibilities based on their individual strengths and leveraging their clinical duties to aid in customer discovery interviews. PI's and team members strategized which faculty or other critical stakeholders they would likely encounter over the next 8-weeks and how they might engage in interviews or feedback of their prototype and solution. Utilizing this information, teams created a customer discovery strategy and interview guides.

TABLE 2. Two-Hour in Person Kickoff Workshop

Topic	Content and/or Activity	Time
Introductions	All individuals introduce themselves and briefly describe their project and/or role within the team.	10 minutes
Expectations	CSI leadership: Provide coaching, resources, accountability, communication, and availability. Participants: Communication, good faith effort, program completion, teamwork.	10 minutes
Scrum overview & STAIR framework	-Scrum theory & healthcare case study examples.[1-3] -STAIR framework.	15 minutes
Customer discovery	-What is customer discovery and why is it important? -Customer discovery planning. -All teams create interview guides and stakeholder interview plans.	30 minutes
Team communication	Set expectations for team communication frequency and discuss preferred communication style.	10 minutes
Prototyping	-Getting started with prototypes. How residents can work with engineers. -Works-Like vs Looks-Like. -Translating user needs to design. -STAIR framework will use prototypes to aid customer discovery. -Engineering consult about initial prototype ideation (supplies, design, feasibility).	45 minutes

Completion of this workshop signified the start of the 8-week course.

The CSI provided resources and guidance for teams to navigate their initial (3) *Prototyping* design and creation. Teams were allowed 24/7 access to the 500 sq ft CSI Innovation Lab which enables collaborative workspace and is supplied with work benches and basic hand tools. The University of Michigan Fabrication Underground studio was also a resource for teams. This studio provides U-M community access to resources including work surfaces, hand tools, 3D printers, and electronics. PIs were responsible for communicating their vision and co-creating the design with engineering support. Most teams dedicated 10-14 days to prototype ideation, design, and development. PI's provided daily guidance and feedback for engineering support, typically in the form of asynchronous communication, brief phone, or video conference. Prototypes ranged from a sketched drawing, computer aided design, 3D printed model, or supplies to be purchased or repurposed to build a physical model. Each team was given a $500 budget for prototyping supplies. Prototypes were purposely non-functioning and not to scale. For Team 1, it was determined that "any amount of light for any given time" would be an improvement to current standards in tight enclosed procedures. The team created a basic prototype; mounting lights onto a commonly used retractor and simulating difficult visualization with a simple tube as an analog for the size and shape of a surgical site. The team shared their design concept using battery powered LED lights with identified stakeholders during (4) *Daily Scrum & Customer Discovery*.

Teams dedicated exactly one week to conducting daily interviews with peers and faculty members. PI's pre-assigned identified stakeholders for team members to interview. Interviews were strategized based on clinic schedules and team members proximity to an identified stakeholder. In person, 5-10-minute interviews were preferred; some teams arranged 15+ minute virtual or in person interviews. All team members and a CSI leadership team member participated in daily communication with one another to report high-level findings and obstacles. Residents triaged obstacles and assigned new tasks for the next day.

After one week of daily interviews, teams transitioned to (5) *Refine* where they shared in-depth findings and feedback with CSI leadership about data acquired during (4) *Daily Scrum & Customer Discovery*. After thoroughly discussing data and feedback, teams decided whether to proceed with repeating steps (3-5) or move toward step "(6) *Definition of Done*." All teams completed steps (3-5) within 4 weeks and repeated these steps for the remaining 8 weeks of the course. Team 1 used feedback collected during their first round of (3-5) to pivot from a retractor mounted light source to a more novel design they aim to patent; they utilized $237 of their budget on supplies.

TABLE 3. Surgical Innovation Discovery Course Final Presentation Slide Template

Slide 1	**Team introductions**
Slide 2	**Unmet surgical need:** *Describe problem or current standard of care, etc.*
Slide 3	**Solution:** *Explain your solution to this problem.*
Slide 4	**Getting started:** *Describe initial steps to prepare for exploring a solution to this problem.*
Slide 5	**Assumptions:** *Before you started work on this project, what did you assume to be true? Who were the stakeholders you thought were important or directly impacted by this problem?*
Slide 6	**Customer discovery:** *Who did you interview? What questions did you ask? What did you learn? How did your assumptions hold true/pivot/evolve during your interviews?*
Slide 7	**Customer discovery 2.0:** *Who would you like to interview again? What would your future approach be to handling interviews? Who else would you like to interview in the future?*
Slide 8	**Prototype:** *How did you prepare for your prototype? What did you learn? What feedback did you receive? What modifications are needed for the next iteration? Show pictures / display in person.*
Slide 9	**Next steps:** *If you were to continue working on this idea, what would your next steps look like? What do you need to be successful? If you are no longer interested in working on this project, why? Is it able to be handed off to someone else?*
Slide 10	**Key takeaways:** *What was this experience like for you? What did you learn? What was challenging?*

"(6) *Definition of Done*" is final step of the STAIR framework and reflects the team acknowledging that they have thoroughly explored their idea. Teams then decide to continue with additional innovation programming support or cease work on the project. At the formal conclusion of the course, all teams gathered in person to report their progress and potential next steps. PIs selected this date 90-days in advance. A final presentation slide deck template was provided to all teams (Table 3).

CONCLUSION

The STAIR framework created a format to empower surgical trainees to form and lead innovation teams to explore solutions to clinical problems utilizing a team-based fast-fail approach. Providing necessary infrastructure (resources, mentorship, and team members) was critical to the successful implementation of this framework. Surgical trainees can continue to utilize this lean-innovation framework throughout their training and careers to lead teams to explore early-stage innovative solutions to clinical problems.

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